

Dhanush Kumar Reddy Pothala | Data Scientist

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SUMMARY

Data Scientist with **3+ years of experience** designing predictive, NLP, and LLM models that enhance decision accuracy and reduce operational costs. Proficient in Python, PySpark, AWS SageMaker, and MLflow for large-scale deployment. Skilled in fraud detection, churn analysis, and model explainability using SHAP and LIME. Experienced in automating pipelines and creating actionable insights with Tableau, Power BI, and Snowflake.

TECHNICAL SKILLS

Programming & Scripting: Python, R, SQL, Scala

Data Analysis: Pandas, NumPy, PySpark, Dask, Feature Engineering, Data Wrangling, EDA, Statistical Modeling

Machine Learning: Scikit-learn, XGBoost, LightGBM, Cat Boost, TensorFlow, PyTorch, Keras, Model Development, Optimization, Explainability (SHAP, LIME)

Deep Learning & AI: Neural Networks, Transformers, NLP (BERT, GPT, Hugging Face, LangChain), Computer Vision (OpenCV, YOLO), Generative AI (GANs, Diffusion Models)

Big Data & Distributed Systems: Apache Spark, Databricks, Hadoop, Kafka, Hive, Snowflake, BigQuery, Delta Lake

Data Visualization: Tableau, Power BI, Looker, Plotly, Matplotlib, Seaborn, Dash, Streamlit

Cloud & MLOps: AWS (S3, SageMaker, Glue, Redshift), Azure ML, GCP Vertex AI, Docker, Kubernetes, Airflow, MLflow, Kubeflow, CI/CD, Flask, FastAPI, Terraform

Databases & Warehousing: MySQL, PostgreSQL, SQL Server, MongoDB, Snowflake, Redshift, BigQuery, ETL/ELT, Data Modeling

Statistics & Mathematics: Regression, Classification, Clustering, Forecasting (ARIMA, Prophet), Hypothesis Testing, A/B Testing, Dimensionality Reduction

Business & Analytics: Predictive Analytics, Forecasting, Risk Modeling, Experiment Design, KPI Measurement, ROI Analysis

Governance & Compliance: Data Quality, Privacy (HIPAA, GDPR, CCPA), Model Monitoring, Responsible AI

Collaboration Tools: Git, GitHub, JIRA, Confluence, Agile, Scrum

PROFESSIONAL EXPERIENCE

Mastercard, USA

Jan 2025 - Current

Data Scientist

- Developed and deployed fraud detection models using PyTorch and XGBoost, reducing false positives by 31% and saving over \$2M annually across global risk systems.
- Systematized data ingestion and transformation for 100M+ daily transactions with PySpark, Airflow, and AWS Glue, improving pipeline efficiency by 85%.
- Integrated LLM-based fraud summaries using GPT and LangChain, generating mechanized case narratives that cut analyst review time by 22%.
- Enhanced model transparency and compliance using SHAP and LIME, ensuring alignment with GDPR, CCPA, and Mastercard's Responsible AI standards.
- Accelerated model deployment from 3 weeks to 5 days through AWS SageMaker, MLflow, and automated CI/CD workflows, improving release frequency.
- Designed interactive Power BI and Tableau dashboards visualizing fraud KPIs, model drift, and precision trends for executive-level decision-making.
- Collaborated with Risk, Engineering, and Product teams to refine fraud detection logic, improving transaction accuracy and reducing chargeback losses by 17%.

Cognizant, India

Aug 2021 - Jul 2023

Data Scientist

- Improved customer retention by 18% and conversion by 12% through predictive churn and lead scoring models using LightGBM, Cat Boost, and Scikit-learn.
- Engineered Spark and Snowflake pipelines processing 2TB+ data daily, reducing reporting delays from six hours to under two.
- Increased model reliability by 25% through advanced feature engineering, EDA, and hypothesis testing with Python, R, and Pandas.
- Enhanced forecast accuracy by 14% by deploying ARIMA and Prophet models to optimize campaign spending and resource allocation.
- Improved A/B test precision by 21% using statistical validation frameworks and confidence interval modeling to guide product decisions.
- Deployed scalable ML models on Azure ML and Kubernetes, ensuring 99.9% uptime and computerized model retraining for live applications.
- Delivered actionable insights through Tableau and Plotly dashboards, helping leadership identify performance trends and strategic opportunities across campaigns.

EDUCATION

Master of Science, Computer Science | University of Houston-Clear Lake, Houston, Texas

May 2025

Bachelor of Technology in Computer Science | Vellore Institute of Technology, Vellore, India

Jun 2023

PROJECTS

- Clinical Readmission Risk Prediction System (Healthcare | AWS):** Developed a HIPAA-compliant ML model using Python, SQL, and AWS SageMaker to predict 30-day patient readmission risk (AUC = 0.82), improving triage accuracy by 48% and reducing manual effort by 35%.
- Solar Asset Predictive Maintenance & Energy Analytics (Energy | Azure):** Built time-series models in Azure ML using Python, SQL, and Data Factory to predict solar equipment failures (precision = 0.76), cutting downtime by 20% and improving energy yield by 4%.